

# Medical Electronics

## Semiconductors Driving Innovation

**In an emerging market like India, there is a lot of potential for growth and demand for medical electronics. And semiconductors are set to play a pivotal role in designing and developing portable and affordable medical devices**



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**M**edical electronics is the convergence of electronics with the realm of medicine. Every time you visit your doctor or the hospital for medical care, you benefit from medical electronics that has become a part of healthcare. From measuring your blood pressure using the tiny electronic gadget you bought from the pharmacy, getting a diagnostic scan done by your doctor, up to a major surgery at the hospital,

there are semiconductor chips working hard to enable such medical advancements. Medical electronic gadgets like scanners, blood pressure monitors and ultrasound systems are essential for diagnostics and perform a critical function in determining the right treatment. The most complex surgeries use state-of-the-art electronics for imaging and robotics.

Taking medical electronics to the next stage is wireless technology, which is now almost omnipresent with the spread of mobile phones in India.

This enables medical care to go mobile as well with the integration of 3G wireless modules in diagnostic devices like cardiac monitors for heart patients and glucose meters for diabetic patients. These electronic devices allow real-time monitoring by the physicians and fast and timely corrective action. Individuals are empowered to monitor their own health, make adjustments based on the data and get required medical assistance before the situation becomes serious. This is significant for patients requiring remote or home-based care, like senior family members at their homes.

The medical electronics ecosystem comprises original equipment manufacturers (OEMs), system integrators, independent design houses for software and hardware development, semiconductor manufacturing houses, mechanical components manufacturing houses, etc.

OEMs include majors like GE, Larsen & Toubro, BPL and Siemens. Solution providers include the likes of Texas Instruments, Freescale, Xilinx, STMicroelectronics, Microchip, MindTree and Magma Design Automation. Besides, the Department of IT, government of India, is developing technologies involving considerable investments and providing them to the industry for commercialisation.

### Technological trends

To understand the reach and impact medical electronics is having on the patient care, consider a few cases: Sur-